TECHNOLOGY

The Brochure Series of Architectural Illustration, Vol. 01, No. 02, February 1895. Byzantine-Romanesque

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[Illustration: IX. The Principal Doorway to the Cathedral at Trani, Italy.]

THE BROCHURE SERIES

OF ARCHITECTURAL ILLUSTRATION.

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BYZANTINE-ROMANESQUE DOORWAYS IN SOUTHERN ITALY.

The illustrations chosen for this issue are all from the Byzantine Romanesque work in the province of Apulia, that portion of Southern Italy familiar in school-boy memory as the heel of the boot. Writers upon architecture have found it difficult to strictly classify the buildings of this neighborhood, as in fact is the case with most of the medieval architecture of Italy, although the influences which have brought about the conditions here seen are in the main plainly evident. The traditions and surroundings, of Roman origin, were modified by trade and association with the Levant through the commerce of Venice and Pisa, resulting in a style embodying many of the characteristics of both the Romans and the builders of Byzantium. Oftentimes these characteristics are so blended and modified by one another as to be entirely indistinguishable, while at other times features unquestionably belonging to the Romanesque or the Byzantine will be found side by side. An illustration of the latter condition may be seen in the two views of the doorway to the cathedral of Trani. (Plates IX. and X.) On account of the intimate relations maintained during the Middle Ages between this province and Magna Grecia, and it may be partly on account of the comparative remoteness from the principal cities of the north, the Byzantine influence is here more strongly marked than in the cities of Central and Northern Italy.

According to the classification adopted by Fergusson, the church of San Miniato at Florence is one of the oldest examples and a good type of this rather mixed style. It was built about the year 1013. It is rectangular in plan, nearly three times as long as wide, with a semicircular apse. Internally it is divided longitudinally into aisles, and transversely into three nearly square compartments by clustered piers, supporting two great arches which run up to the roof. The whole of the inner compartment is occupied by a crypt or under church open to the nave, above which is the choir and altar niche, approached by flights of steps in the aisles. This general arrangement is followed more or less closely in the churches at Bittonto, Bari, Altamura, Ruvo, Galatina, Brindisi, and Barletta. The scale of the southern churches is, however, much smaller than those of the north, the width of the nave of the cathedral at Trani being only 50 feet, and the length 167 feet, while the corresponding dimensions of the cathedral at Pisa, which is referred to by Fergusson as the most notable example of this style in the north, are 106 x 310 feet.

In these smaller churches, as far as external treatment is concerned, the main attention is devoted to the principal façade, and here most of the ornament is usually covered with a rich hood supported by pillars resting on monsters, following the custom prevalent throughout Italy during this period. Above this is either a gallery or one or two windows, and the whole generally terminates in a circular rose window filled with tracery.

[Illustration: X. The Principal Doorway to the Catherdral at Trani, Italy.]

Fergusson's final summing up of the architecture of this neighborhood can scarcely be considered too enthusiastic in the light of the eight illustrations here given. He says: "No one who takes the pains to familiarize himself with the architecture of these Southern Italian churches can well fail to be impressed with their beauty. That beauty will be found, however, to arise not so much from the dimensions or arrangement of their plans, or the form of their outline, as from the grace and elegance of their details. Every feature displays the feeling of an elegant and refined people, who demanded decoration as a necessity, though they were incapable of rising to any great architectural conception. They excelled as ornamentists, though at best only indifferent architects."

The examples of doorways chosen for illustrating this number unquestionably show the work of men who labored for the enjoyment and satisfaction to be got from their work. This is sufficiently evident in the results before us. Its logical and constructive bearing can of course be called in question, as in fact is the case with all but the merest fraction of the architectural efforts of the world. As decoration we can but admire the masterly way in which the ornament is distributed, the refined sense of scale and proportion, and the skilful and subtle treatment of light and shade, even if the detail of the ornament itself is crude and archaic.

In making the choice of these subjects this point was kept in mind, and they are not offered as material which can be cut out in portions of the size and shape desired and transferred bodily by the designer to embellish a modern masterpiece, in the manner in which the Gothic architects of Venice used their patterns of window tracery. These plates show certain qualities in decorative design in their fullest and best development, and are on this account invaluable as suggestions to designers of the present day. For "cribbing material" they do not stand for much; but this should not be counted as against their usefulness, for the draughtsman who has not advanced beyond the "cribbing" stage has much still to learn before he can do the best and most satisfactory work.

IX. and X.

PRINCIPAL DOORWAY TO THE CATHEDRAL AT TRANI, ITALY.

The cathedral at Trani dates from about the middle of the twelfth century. Its main features have been indicated above in describing the general characteristics of the class of churches to which it belongs. The bronze doors shown in the illustration were made in 1160, and are exceptionally fine examples of the work of this period.

XI.

PRINCIPAL DOORWAY TO THE CATHEDRAL AT CONVERSANO, ITALY.

Doorways of this general design are so familiar in the so-called Romanesque architecture of our American cities that it seems almost like an old friend; but we regret to say that most of our American designs would hardly show to advantage if compared side by side with this.

XII.

PORTION OF THE FACADE OF THE BASILICA AT ALTAMURA, ITALY.

The remarkable sense of spotting and distribution of ornament shown in the designing of this facade can hardly be too much commended. The strong light and long slanting shadows of the photograph are well calculated to emphasize this quality in the design, and we can readily find justification here for the estimate of Fergusson quoted above.

XIII. and XIV.

PRINCIPAL DOORWAY TO THE BASILICA AT ALTAMURA, ITALY, AND DETAIL OF THE SAME.

XV.

DOOR OF MADONNA DI LORETO, TRANI, ITALY.

XVI.

ENTRANCE TO THE CHURCH OF THE ROSARY, TERLIZZI, ITALY.

[Illustration: XI. The Principal Doorway to the Cathedral at Conversano, Italy.]

#Advice to Young Architects.#

Prof. Aitchison's Royal Academy Lectures upon Architecture should be read by all students who can obtain access to them, and this is not really very difficult to accomplish, as they are always reported at length in the English architectural periodicals, and then usually reprinted without credit by one or more of the American papers. The latest one, reported in the _Builder_ of Feb. 16, is that delivered on Feb. 4, under the general title "The Advancement of Architecture." It deals in a common-sense fashion with the aesthetics of architecture, and contains many valuable suggestions upon the study and practice of architecture as an art. The three following quotations are well worth attentive reading:--

"Swift, in his 'Letters to a Young Clergyman,' says: 'I cannot forbear warning you in the most earnest manner against endeavoring at wit in your sermons, because, by the strictest computation, it is very near a million to one that you have none.' Perhaps that would be good advice to all who consciously seek for what is called originality,

which is mostly attained by exaggeration, disproportion, and oddness of arrangement; real originality only comes from original minds, and will in that case show itself properly and naturally, just as wit shows itself spontaneously in the witty; for surely those original architects, who have only been able to raise in us emotions of contempt or disgust, would have been judicious had they abstained from the attempt. I think that most architectural students, if they will only study the best buildings, will make their plans to accurately answer the purposes wanted, including the efficient lighting of the rooms, will study the Vitruvian symmetry until their eye revolts from disproportion, will try and make their profiles tell the story they want told, and will try and bring such parts that, from the exigencies of the case, obtrude themselves in odd places into harmony with the whole, that they will produce an effect which will raise their buildings to the dignity of humanity, and out of the range of the dog-kennel and rabbit-hutch type, and will not exhibit ugliness, disproportion, or vulgarity. We see plenty of examples where the designs have sunk much below this level; no building of dead walls, with holes in it for doors and windows, could cause us such disgust. Let me here say, by way of a parenthesis, that if you candidly consider that your design is more offensive than a dead wall, do not waste money and materials in making the wall more repulsive, but let it alone."

"Any one can be original if he be only impudent enough; any one can be graceful if he is servile enough to copy: but to be both original and graceful requires deep study, much striving, and natural talent."

"I have also to remind you that architecture cannot be brought into vigorous life again, so long as architects insist on using old forms for beauty that are inseparable from a construction that has been abandoned; so long as this practice persists, so long will architecture be a kind of potted art; to be vigorous it must learn how to take the materials, and construction that would be ordinarily used in buildings for purely practical purposes, and give to these materials and this construction forms that will excite the proper emotions. You must not suppose that I mean that if you have a vast hall, or what not, that because you can put an iron trussed roof over it from wall to wall, that this will make it into a hall that will raise emotions. You will only get a rail-way platform or a coal shed. You have got to set your wits to work to see how it can be properly brought within the pale of aesthetics, and not only as to the shapes and proportions of the parts, but the dividing of the whole by supports. It is probable that if you were obliged to vault a cathedral in stone, with no more money than was necessary, and to have a clearstory to it, that you could not do it cheaper, and perhaps not better, than the Gothic architects did it; but to vault such a building in stone when you could do it much cheaper and better with iron ribs and concrete is, in my opinion, _dilettante_ art. Groins are not beautiful things, but, on the contrary, are ugly, and we should wish to obviate their ugliness if we could; but when they were merely unavoidable methods of cheap construction, we admire them for the invention and skill of their architects, and we have to some extent got to love even their ugliness from old association; though perhaps the ribs at Westminster Abbey, as seen from the west end, are not offensive."

[Illustration: XII. A Portion of the Façade of the Basilica at Altamura, Italy.]

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All who wish for a complete file of THE BROCHURE SERIES should send in their subscriptions at once, as owing to the necessity of limiting the edition of the first numbers and the impossibity of reprinting when this edition is exhausted, subscriptions will have to date from the current number at the time the order is received. Until the present stock gives out, all subscriptions will be dated from the January number, but no copies will be reserved for this purpose after April 1.

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Response to the call for subscriptions to THE BROCHURE SERIES has been gratifyingly prompt and generous. The first subscriber was Mr. George B. Howe, 13 Walnut Street, Boston, the architect of the New Hampshire State Building at the World's Fair. The first club came from the office of Longfellow, Alden & Harlow, and was made up as follows: F.B. Wheaton, R.T. Walker, H.W. Gardner, H.M. Seaver, and J.H. Buttimer. This was closely followed by a club of eight from the office of Shepley, Rutan & Coolidge, and another of five from the office of Edwin J. Lewis. The first response from out of town was a club of five from the office of Martin & Hall of Providence, R.I. Others "too numerous to mention" came along in quick succession, and the new magazine may now be considered well launched on its trial trip.

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As the plan of THE BROCHURE SERIES is unique in architectural journalism, much of the work to be done during its first year will necessarily be, to a certain extent, experimental. Although the publishers have for a number of years tried to keep as closely as possible in touch with the profession throughout the country, the diversity of tastes to which the new magazine is intended to appeal, and the practical requirements which it is intended to meet, make even the simple matter of selecting proper material for publication a difficult task. Consequently suggestions or criticisms which may lead to its improvement in any particular will be welcome.

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The design used for the cover of THE BROCHURE SERIES is the result of a competition in which twenty-three drawings were submitted, and is the work of Charles Edward Hooper of 250 West 14th Street, New York. The other competitors,

whose designs were all of a high order of excellence, were: J. Mills Platt, Charles S. King, Francis S. Swales, Edwin S. Gordon, Fred A. Miller, J.F. Strobel, Jr., George E. Roberts, of Rochester, N.Y.; G.H. Ingraham, E.P. Dana, F.H. Hutchins, C.E. Patch, of Boston; J.W. Cinder, W.B. Papin, H.G. Helmerichs, of St. Louis; Louis E. La Baume, H.H. Braun, of New York City; and Stephen W. Dodge, of Brooklyn.

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Following out the general plan adopted in the first two issues, which, contrary to the expectation of the publishers, has proved even so soon an important feature of the magazine, the illustrations in the next two numbers will be made up of related subjects. The March number will have a collection of capitals (Byzantine and Romanesque) from Ravenna and Palermo, and the April number eight windows from Apulia, of the same general character as the doorways in the present number.

#Hints to Draughtsmen.#

Architectural students and draughtsmen will find the series of papers begun in the Feb. 16 number of the _American Architect_, entitled "Hints to Art Students on Travelling Abroad," filled with valuable suggestions. The writer of these papers is Mr. J.W. Case, the latest of the Rotch scholars returned. In the first paper Mr. Case points out the desirability of preparatory training in academic design, drawing, modelling, etc., and a knowledge of architectural history and of the French language in order that the student may make the best use of the opportunities open to him. He continues with a number of useful hints upon the best methods to pursue in gaining this preparatory training.

[Illustration: XIII. The Principal Doorway to the Basilica at Altamura, Italy.]

The second paper is devoted to practical suggestions of such immediate value that it is worth while to quote a portion of them in full:--

"To get the most good out of a trip, one should be prepared to work in all sorts of ways,--to make measured drawings, sketches, color notes, squeezes, rubbings, sections with the lead; to study from plates and make T-square sketches, scratch-book notes, photographic notes, and memory sketches.

"Travelling students are apt to place too much value on perspective sketches. Good ones make a nice showing on returning home, but they are of little value to any one but the maker. It is usually possible to find photographs of the things over which one spends so many hours making pretty sketches. But sketches do have a certain value in teaching rendering, and encourage the habit of observing closely the effect of light and shade.

"Beautiful pencil sketches may be made on English metallic paper by simply drawing the shadows on carving in full sunshine: colored papers are very useful to gain quick effects with the use of Chinese white. A pad of Whatman water-color paper, imperial size, is much better to work on than a small cramped little book; and it may be used as a drawing-board, thus diminishing the number of articles to carry. The T-square will run along the edge of the block well enough for sketches, but it is better to carry a straight-edge to clamp on the edge of the block with thumb-screws

for the square to work on. Have a canvas bag made with a flap in which to carry the block. It will keep out the dirt and dust of travel and be of great service.

"Sometimes valuable color notes are to be had in crowded buildings where it is not convenient to sit down and make a large study. For such cases a small pocket water-color block will be very useful. There is a small vest-pocket water-color box carrying six colors, which may be set over the thumb, a water-bottle attached, and with it one can stand unobserved in a corner and get color notes which otherwise must be passed by. In studying fresco painting, tempera is very useful. It is mixed up with water and applied to paper, but may be worked over in the manner of oils,--a great advantage in making studies.

"The _chambre éclaire_ is invaluable as an aid to drawing, in blocking out water-colors. It will enable one to make a drawing in an hour which otherwise would require all day. It is an instrument little known outside of Paris, but is much in use there among architects. It consists of a prism mounted on a telescoping leg which may be fastened to the drawing-board. The eye looks through the prism and sees the building reflected on the paper; all that remains to do is to trace this outline. It does not teach one to draw, but it does save time, and produces better drawings than can be made without it. The best place to buy them is of Cevalier, on the Seine, near the Pont Netif, Paris. Only those with the best prisms are of any use: such a one, with two adjustments only, can be had for sixty-five francs. The table which is necessary for its use costs fifteen francs additional; that is, a total cost of sixteen dollars. In buying a table, be sure and get one with sliding legs which can be taken off the head and packed flat.

"One of the very best ways to study, and one which has very direct tangible results, is by the aid of printed plates. Take such a book as Letarouilly's _Edifices de Rome Moderne_. Go to the buildings themselves and compare the drawing with the building; see what drawings on paper really mean when executed; mark up the plate; note the proportion of masses, the size of ornament, the relative proportion of openings, and wall spaces, the effect of color and texture, and the use of material. Make suggestions for better ornament, proportion, etc., and then go home and make a new design with all the improvements you have noted.

"The reverse of this method is, to sit down in front of the building with T-square and triangle and translate the perspective building back on to paper in elevation.

[Illustration: XIV. Detail of the Principal Doorway to the Basilica at Altamura, Italy.]

"These two methods will aid one to tell from a drawing how the building will actually look when executed. It will give an idea of the scale of ornament, if a cornice looks just the right size on a certain building, the plate will tell you just how high that is. The T-square sketch is very valuable in cultivating the sense of proportion. Draw to scale such parts of the sketch as can be easily measured, and put in the remainder in proportion, and make these sketches at the scale at which you are used to working in the office. They will be of immense advantage in giving you a sense of absolute scale.

"There is such a thing as 'absolute scale,' and scale is not simply proportion. A drawing might be made in good proportion, and the building look well if executed a thousand feet long, and yet lose all its effectiveness if executed but one hundred feet in length, the relative proportions of the parts remaining the same. It is a fact that certain designs, which look well on paper, will not look well in execution, except at a large scale. Therefore it is valuable in making a sketch to put on it some of the measurements; and freehand sketches with measurements marked on them have a value in giving absolute scale.

"The back of a photograph is a very convenient place on which to make notes of the building itself, in regard to color, material, suggested changes, etc., and will be very useful in recalling the building to memory.

"Measuring buildings and drawing them out to scale is solid architectural work, and nothing else can take its place. It gives a realization of the actual size and appearance of things, and brings to notice the stone-jointing, sections of mouldings, vaulting, roofing, and construction in general. Measured work must be done very accurately, or else the results have no more value than approximate measures on sketches.

"The drawing should be made exactly as the building exists, without any change or improvement, or else the drawing will lose a great deal of its value as a basis for study. Many of Letarouilly's are nearly valueless as data for study because he has improved on the original, and thus his drawing does not represent the building as it actually exists.

"A good method of measuring buildings is to measure first the general dimensions and block out the building on paper at a small scale, then measure up windows, columns, etc., and set off full-size sections of all the mouldings with a strip of thin lead, such as may be had at any whole-sale lead store: only the thinnest sheet-lead will work, as the thicker leads are too stiff to bend. The large final drawings can then be made away from the building. It is important to draw out the building completely at a small scale, however, as it is very annoying when making the final drawing far away from the building to find that some important dimension has been forgotten.

"The ordinary tape stretches so much in long dimensions that it is inaccurate. It is best to get a tape with a metallic strip in it, and it should be at least fifty feet long in order to take dimensions over all, which is much more accurate than measuring with a short tape from point to point.

"The metric system is very convenient, but it is better for American students to use the English measure that they will have to use in practice, and take the tape over with them, for it is difficult to find them on the Continent. A sliding measuring-rod is nearly indispensable, and it will be most convenient to carry if it folds up to the length of the imperial drawing pad. Two large triangles are very useful in getting the projection of mouldings, as they can be held together to form a right angle."

[Illustration: XV. Door of the Madonna di Loreto, Triani, Italy.]

#Books.#

Verona and Other Lectures. By John Ruskin, D.C.L., LL.D. New York: Macmillan & Co., 1894. 8vo, pp. 204, plates xii. $2.50.

The art of Northern Italy has furnished the text for a very considerable part of the writings of Mr. Ruskin, and there is no one writer among those who have ventured to investigate and write upon this extremely engrossing subject whose work has so great an interest for the architect, or in fact is of so much value to him. It is not necessary to agree with all of Mr. Ruskin's elaborate theories or to unqualifiedly admire his drawings in order to find much of real value in his books. No student of architecture can afford _not_ to read "The Stones of Venice," and there are few books which should take precedence over it in the formation of an architect's library.

Apropos of the illustrations in the last number of THE BROCHURE SERIES, in the descriptive notices of which we had occasion to refer to Mr. Ruskin, his latest published work will be found interesting. The title, "_Verona and other Lectures_," does not convey a very complete idea of the contents of the book. None of the five lectures included is strictly architectural in subject matter, and but one, the first, "Verona and its Rivers," has any direct bearing upon architecture, and this only from the historical side. The illustrations, with a single exception from drawings by the author, although lacking in most of the qualities of good draughtsmanship, are well worth examination and study. Plates II. and V., "A Fountain at Verona," and "The Castelbarco Tomb, Sta. Anastasia, Verona," the first made in 1841 and the second in 1835, are from the point of view of the architect the most interesting. They are both pencil sketches, the first accented with a few touches of wash in the shadows and darker portions of the drawing. Plate IX. represents the angle of the Ducal Palace, Venice, the same given as the frontispiece in the last issue of THE BROCHURE SERIES. It would hardly be possible to come nearer the same point of view if the coincidence were intentional. In the comparison which this forces upon Mr. Ruskin very naturally suffers, as might be expected, from the fact that his training in drawing was not the most thorough. His proportions are somewhat faulty and the detail is only vaguely suggested, in fact this is more or less true of all his drawings. Nevertheless the book will be welcome to many architects for the valuable suggestions it contains both in text and illustrations; and the author's wonderful and fascinating literary style is here as unmistakably in evidence as in any of his older works. This alone is sufficient inducement to tempt the reader to take it up.

#Club Notes.#

At the suggestion of several subscribers, the addresses are given below of the secretaries of the principal architectural clubs as far as they are known to us, but there are several omissions and possibly some mistakes. In order that these associations may be of as great mutual assistance to each other as possible, through correspondence, the exchange of notices of competitions, etc., it is requested that any not included in the following list will communicate the desired information to the editor of THE BROCHURE SERIES. Corrections or additions will be made in later issues, and the various secretaries will confer a favor by keeping the editor informed of any changes of address or organization.

LIST OF CLUBS.

Sketch Club of New York, club rooms 1473 Broadway; recording secretary, Alfred F. Evans; corresponding secretary, Hobart A. Walker.

Boston Architectural Club, rooms 5 Tremont Place; secretary, F. Manton Wakefield.

The T-Square Club, Philadelphia, rooms Broad and Pine Streets; secretary, A.C. Munoz, 212 South Third Street.

Chicago Architectural Club, rooms 274 Michigan Avenue; secretary, John Robert Dillon.

St. Louis Sketch Club; secretary, E.G. Garden, Telephone Building.

Art League, Milwaukee, Wis.; secretary, Elmer Grey, 904 Winchester Street.

St. Paul Architectural Sketch Club, rooms 239-241 Endicott Building; secretary, John Rachac, Jr.

Cleveland Architectural Club, rooms 1002 Garfield Building; secretary, Herbert B. Briggs.

Denver Architectural Sketch Club; president, William Cowe, 706 Cooper Building.

Rochester Sketch Club, secretary, G.F. Crump, Wilder Building.

The Architectural League of New York, American Fine Arts Society Building; secretary, Charles I. Berg, 10 West 23d Street.

The Society of Beaux Arts Architects. New York City.

[Illustration: XVI. Entrance to the Church of the Rosary, Terlizzi, Italy.]